

Nonnegative and positive factorizations

# The nonnegative rank of the nine-point distance matrix

**Difficulty:** hard **Importance:** interesting to specialist

**Status:** Open

**Rating rationale:** Hard because a single small exact factorization must be constructed or excluded using sharp nonnegative-rank tools; specialist importance reflects its role as a named distance-matrix benchmark.

**Area:** exact factorization of Euclidean distance matrices

## Context and notation

All factorizations are over the real numbers. For  $X \in \mathbb{R}_{\geq 0}^{m \times n}$ , define

$$\text{rank}_+(X) = \min\{r \geq 0 : X = WH, \quad W \in \mathbb{R}_{\geq 0}^{m \times r}, H \in \mathbb{R}_{\geq 0}^{r \times n}\}.$$

## Problem statement

Let  $D \in \mathbb{R}_{\geq 0}^{9 \times 9}$  have entries  $D_{ij} = (i - j)^2$  for  $1 \leq i, j \leq 9$ .

## Question

Does there exist an exact factorization  $D = WH$  with  $W \in \mathbb{R}_{\geq 0}^{9 \times 6}$  and  $H \in \mathbb{R}_{\geq 0}^{6 \times 9}$ ? Equivalently, is  $\text{rank}_+(D) = 6$  or  $7$ ?

## References

Baeckelant, Vandaele, and Gillis, *Computing Lower Bounds on the Nonnegative Rank via Non-Convex Optimization Solvers*, Appendix A.1, Table 6 and its final paragraph, expressly isolates this case. Pavel Hrubeš, *On the nonnegative rank of distance matrices*, Information Processing Letters **112** (2012), 457–461, supplies the upper-bound construction cited there.

## Status check — 2026-09-10

Rechecked [Baeckelant et al. v2, Appendix A.1, Table 6 and final paragraph](#), and searched for later nine-point distance-matrix factorizations. The July 2026 text explicitly retains the gap between six and seven. Its new exact results for orders eleven and twelve do not settle order nine. No six-factor construction or seven-factor lower bound was located.